



› FOR SOCIETY

SOUND SOURCE LOCALIZATION

ANALYSIS DOCUMENT

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VERSION HISTORY

Version	Date	Author	Changes
1.0	12-04-2024	Tom Verlinde, Yassine Tama, Casey Stijlaart, Teun Spithoven	Initial version

INTRODUCTION

This document was converted from MD to Typst. The original text is present on the team wiki

BRIEF OVERVIEW

Within Fontys (and other companies) a new type working environment was introduced, the open office or OIL (Open ICT Lab) within Fontys. In these spaces people can move around freely and aren't confined to one room/desk. One big downside of these open environments is that it naturally creates lots of noise which leads to distractions and frustration. This project is an attempt to combat this.

The project previous goal was to produce a system that could monitor the actual level of noise coming from one spot, and to actually locate where the sound is coming from using triangulation. This way people can be notified when they are being too loud, as opposed to the previous version where the sound wasn't triangulated.

PURPOSE OF THIS DOCUMENT

The purpose of this analysis document is to get a good perspective of what the previous group has achieved, where to pick-up the progress, and to look at possible improvements/fixes. Knowing these things will help us with creating tasks, user stories, improvements, but also to get a general feeling for the project.

SCOPE AND OBJECTIVES OF THIS DOCUMENT

This analysis document will essentially look into all the work the previous group did, mainly the documentation which we received from the previous group. The objective is to analyse this and to conclude where we stand right now and what could be improved.

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PROJECT OVERVIEW

As the project is a continuation of a previously worked on project, the previous status and general key factor must be considered. Therefore, a summary of these details are discussed. Such details are the goals/objectives, stakeholders, key deliverables, and a brief timeline with major achievements of the previous group.

OBJECTIVES

The group had a few objectives, that they planned on delivering. These objectives are as following:

- detecting the location of noise
- measuring the noise level
- Visualising the noise in a GUI

STAKEHOLDERS

The stakeholders for this projects are/were:

- Jan Dobbelsteen
- Fontys Dienst Huisvesting & Facilitaire zaken

KEY DELIVERABLES

As there were various aspects were promised on completion of the previous iteration of the project, these are given below, including their status.

- Server with GUI (finished)
- PCB (finished)
- research documents (finished)
- design documents (finished)
- Sound finding algorithm (finished)

TIMELINE AND MAJOR ACHIEVEMENTS

As the previous group worked with scrum, with 3-week sprints, the achievements are given per sprint.

SPRINT 1:

- mainly documentation, setting up the project & research

SPRINT 2:

- research documents were finalised, and conclusions on which technologies to use were made
- multiple POC's were made

SPRINT 3:

- PCB design
- heatmap in GUI
- sound localisation algorithm
- first hardware prototype

SPRINT 4:

- finalised hardware prototype

PREVIOUS PROJECT DEBRIEFING

The project debriefing document of the previous team mostly went into how they set up their project, because of this the document is not all that useful to us. However, there is some information present that might be useful to us.

CONTACT

The previous team has made a table with quite some useful contact information. In this table, they state all people involved with the project, their role, and their contact information.

Name & Email	Role/Task (In the previous semester)
Paul Jans paul.jans@student.fontys.nl	Developer, Communication Manager
Jurgen de Laat J.delaat@student.fontys.nl	Developer, GIT/OneDrive maintainer
Reinier Koffijberg R.koffijberg@student.fontys.nl	Developer
Jesse van de Laar j.vandelaar@student.fontys.nl	Developer
Max van de rijt maxvanderijt.vanderijt@student.fontys.nl	Developer, SCRUM master
Lowie Versleijen l.verleijen@sudent.fontys.nl	Developer
Rens van der Vorst r.vandervorst@fontys.nl	Product Owner
Brice Guayrin b.guayrin@fontys.nl	Field Expert
Jan Dobbelsteen j.dobbelsteen@fontys.nl	Field Expert
Gerard Elbers g.elbers@fontys.nl	Field Expert

FINISHED PRODUCTS

The previous team has listed all the products they claim they have finished.

Product	Description
Project Debriefing	This document contains all necessary project definitions, agreements, and plannings.
System architecture overview	A C4-model based system overview where all components of the system and their purpose are clearly displayed.
Research documentation	Documents where all reasoning behind choices taken during this project is elaborated on.
User interface software	A friendly UI where users can view current noise levels and their locations which are being detected by the system.
Hardware prototype	A prototype consisting of hardware containing the appropriate software.
Server software	The host of the UI and manages and collects data from hardware in the field.
Sprint retrospectives	After every sprint we do a retrospective where we reflect on how the last sprint went and plan out the next sprint.

AREAS OF IMPROVEMENT

Within the current state of the project, there are several key areas that require attention and enhancement. These improvements are based upon suggestions of the previous team as well as our own observations. The identified aspects for improvement are as follows:

PROJECT STRUCTURE

To enhance overall project manageability, a thorough update of the project structure is necessary. This includes revisiting both the git repository organization and the clarity of the source code itself. Currently, the codebase suffers from non-descriptive function naming, leading to confusion for individuals not deeply familiar with the code.

PCB DESIGN

Despite the completion of the initial PCB iteration, several flaws have surfaced during the previous project timeline, necessitating adjustments in the subsequent iterations. The known flaws at the moment are the need to change the design of the power supply, as it seems to be overheating in the current design and the wiring of the GPIO pins that are used for the LEDs also seems to be incorrect. On top of this, a possible improvement could be to move towards SMT instead of THT. This could give the PCB a more “professional look”.

LOCALIZATION

The current localization methodology relies solely on the loudest input from the microphone, which presents an opportunity for refinement. Enhancements could involve incorporating data from all available microphones to improve accuracy. Additionally, optimizing the calculation sequence is essential to mitigate potential performance bottlenecks arising from simultaneous calculations.

TRIANGULATION

While one of the triangulation applications meets acceptable standards, the other requires substantial rework to enhance its reliability and readability. Implementing a configuration file for user-defined settings upon application start-up can significantly improve usability and customization options.

FAST FOURIER TRANSFORM (FFT)

Although the potential benefits of utilizing FFT for the project have been explored, the implementation phase remains pending. It is recommended to incorporate a band-pass filter ranging from 20Hz to 20kHz to exclude irrelevant data and focus solely on frequencies within the human audible range.

WEB APPLICATION

While the current state of the web application is functional, there is room for improvement in terms of user experience and clarity. Enhancements such as integrating microphone and speaker statuses and redesigning control buttons for improved intuitiveness can elevate the user interface and overall usability.

DOCUMENTATION

When looking through the current git repository and the received documentation, it was noticed there was an extreme lack of proper documentation. Various key documents were missing, such as system design document and test plan.

TECHNICAL ANALYSIS

The technical analysis section provides an in-depth examination of the embedded design and overall structure of the project, along with a detailed review of the server architecture and its components. Through meticulous evaluation, critical aspects emerge, shedding light on both strengths and areas for improvement within the system.

The technical analysis consists out of the embedded side and the server side.

EMBEDDED

In examining the embedded design and overall structure, several significant aspects emerge, leading to some critical conclusions. The foremost among these is the presence of structural and consistency issues within the project. These issues hinder a comprehensive overview of the various applications within the system.

For instance, the project comprises multiple applications, yet lacks clear indications of their interconnections and storage mechanisms. Each application is housed within its own folder, containing its source and header files. This organizational approach leads to code duplication and a lack of cohesive management across the applications.

To delve into specific aspects and offer a more detailed analysis, let's consider the following points:

COMMUNICATION PROTOCOL

While a communication protocol is in place, its implementation lacks consistency across the different applications. Each application utilizes its unique message structure, which complicates interoperability and scalability. Moreover, the protocol's shallow depth poses limitations, especially concerning the robustness required for future development and expansion.

TESTING

Regrettably, the project lacks any formal software testing procedures or documentation. There is no evidence of unit testing or integration testing being conducted, which raises concerns about the system's reliability and robustness under various conditions.

DESIGN DOCUMENTATION

Remarkably, there are no discernible traces of existing design artifacts such as class diagrams or sequence diagrams. This absence indicates a significant gap in the documentation of the system's architecture and functionality, hindering comprehensive understanding and future development efforts.

SERVER

The previous team has created multiple web applications, together they act as a centralized server of the sound source localization project. Fortunately, they also made a docker-compose file with all required configuration so it's really easy to run the project and it saves a lot of configuration time.

The server consists of the following applications/components:

- Backend
- Frontend
- User Interface
- Mosquitto message bus
- MySQL database

DATA FLOW

Data from the sensor devices get sent to the message bus via MQTT. The backend receives those messages and stores the readings in the MySQL database. The Frontend and User interface applications read from and write to the backend application.

BACKEND

A Node.js Express API. This is a very common way to make API's so that's great. The API reads from and writes data to the MySQL database. It also listens for messages and sends messages to the message bus.

Unit tests have been made for the API endpoints. They are not yet integrated in the CI/CD pipeline for automated testing.

FRONTEND

The Frontend application is a kind of admin dashboard where all sensor devices and zones can be managed. They are displayed in lists made with barebones HTML, CSS and JavaScript.

The JavaScript is all in one big file at the moment, it can be split into multiple files for better readability and maintainability.

TRIANGULATION

The code for calculating the sound source location based on the angles read by the sensor devices is in this folder under the server folder. It consists of a file without extension (that I can't open) and a python file that gets the sensor data, creates triangles and writes those to a csv file.

Some suggestions for `TriangulateGridMicrophones.py`:

- `retrieve_measuring_units_backend_data()`
 - ✖ The beginning of this function can be fixed by adjusting the API so that you can retrieve data simply by the ID of the measuring unit instead of that strange prefix.
- The long pieces of code for error handling can also be reused to shorten the code.

USER INTERFACE

The User Interface application contains a map of TQ, the sensor devices are displayed on the map with the angle they're placed in. The different zones can also be managed through this interface which is great for usability. This app uses Node.js with bootstrap for styling. Unlike the Frontend application, the JavaScript code is neatly split up into multiple files.

The user experience has room for improvement. It took me some time to figure out how it works, we could make it more straight forward by using more standard UI elements and layouts.

RECOMMENDATIONS & ACTION PLAN

Although a solid foundation has been established for the project, several critical aspects require improvement before integrating new features. Not all mentioned features align with the current objectives of the project and thus won't be prioritized during the current iteration.

KEY FOCUS AREAS

In order to prioritize our efforts effectively, we have identified key focus areas that require immediate attention and improvement. Addressing these aspects will not only enhance the project's functionality but also pave the way for future enhancements and developments.

PCB

The existing PCB falls short of expected standards due to inherent flaws and design deficiencies. Addressing these issues is paramount to improving workflow efficiency and resource utilization.

PROJECT STRUCTURE

Ensuring a coherent project structure is essential for effective collaboration and development. This involves refining the organization of the project within GitLab and enhancing the overall readability and maintainability of the source code.

DOCUMENTATION

As it is of high importance that a project is properly documented, not just for the current group, but for the next group as well. Therefore the missing documentation must be picked up and worked on, to provide a good base for the project.

WEB APPLICATION

Enhancements to the user-friendliness of the web application are crucial. Clear operational guidelines must be provided to ensure all stakeholders can easily navigate and utilize its functionalities.

COMMUNICATION PROTOCOL

While a communication protocol exists, its documentation and implementation lack clarity. Establishing a comprehensive overview and enforcing its adherence are necessary steps to enhance reliability and readability.

TESTING

Implementing a robust testing framework is imperative to validate the functionality of the codebase and ensure consistent performance post-update. Prioritizing the establishment of unit tests integrated into the CI/CD pipeline will facilitate early detection of errors and streamline the development process.

By prioritizing these key focus areas, we aim to bolster the project's reliability, maintainability, and usability, ultimately advancing towards the achievement of our objectives.

CONCLUSION

In conclusion, the continuation of this project builds upon the foundations laid by the previous team, emphasizing the significance of understanding past accomplishments and areas for improvement. Through a comprehensive overview of objectives, stakeholders, key deliverables, and previous achievements, we have established a solid groundwork for our ongoing efforts.

Addressing the identified areas of improvement, including project structure, PCB design, localization techniques, and communication protocols, is vital for enhancing the project's functionality and reliability. By prioritizing these aspects, we aim to streamline development processes, improve workflow efficiency, and elevate the overall quality of the project.

Furthermore, the technical analysis highlights critical considerations in embedded design, server architecture, and user interface development. By addressing issues such as code consistency, documentation gaps, and usability enhancements, we strive to create a more cohesive and user-friendly system.

Moving forward, our action plan focuses on key areas such as PCB refinement, project structuring, web application usability, communication protocols, and testing frameworks. By prioritizing these focus areas, we aim to bolster the project's reliability, maintainability, and usability, ultimately advancing towards the achievement of our objectives.

Once the mentioned issues are addressed, there can be moved forward with adding on to the project.